

SCREEN SENSES: -KIDS' SCREENTIME VISUALIZATION

Milestone 4: Report and Presentation

- Assemble a coherent storyline from visuals
- Build an interactive Tableau or Power BI dashboard with filters (age band, gender, location type)
- Ensure titles, labels, legends, and axis clarity.

1. Introduction

The purpose of Milestone 4 is to convert all insights and visual analyses from Milestones 1–3 into a **structured, interactive, and professionally designed Power BI dashboard**. This stage focuses on **storytelling, visual clarity, and business interpretation**, which are essential for data analyst roles and internship evaluations.

The dashboard is designed across **three pages**, each representing a distinct stage of analysis:

1. **Page 1 – Exploratory Visual Trends (Univariate & Bivariate)**
2. **Page 2 – Device/Activity & Weekday–Weekend Behaviour**
3. **Page 3 – Cohort Segmentation & Deep-Dive Insights**

2. Objective

The main objective is to build a **clear and interactive dashboard** that narrates the complete story of screen-time patterns among children across India.

Specific objectives

- Convert raw visuals into a polished dashboard layout.

- Enhance readability using proper formatting, alignment, and color consistency.
- Add user-controlled filters (slicers) to make the dashboard interactive.
- Combine visuals logically to support storytelling.
- Prepare KPI cards that show core metrics at a glance.
- Ensure accessibility and professional visual standards.

3.Power BI Dashboard Development

Data Import & Power Query Cleaning

In Power Query:

- Standardized column names
- Set correct data types
- Created calculated fields:
 - age_band, age_group, screen_level, is_weekend
- Removed unnecessary columns
- Trimmed whitespace
- Ensured clean modelling for visuals

Designing Visuals

Each chart was built based on best design principles:

- Clear axis labels
- Consistent color palette
- Interactive slicers
- Syncing filters across pages
- Adding titles/subtitles

Visual types used:

- Bar chart
- Column chart
- Area chart
- Scatter plot
- KPI cards
- Heatmaps
- Donut/stacked charts
- Line charts

Adding Interactivity

To make dashboard analyst-friendly:

- Added slicers: Age Band, Gender, Device, Location
- Enabled cross-filter interactions
- Added drill-through to “Cohort page”
- Page navigation buttons

4. Dashboard Design Strategy

A good analytical dashboard must represent a **story**. Therefore, the dashboard design follows a **top-down analytical storytelling** approach:

4.1 Layer 1 – Overview (What is happening?)

Page 1 focuses on distributions, gender differences, device usage, and age-based patterns.

This page answers:

- Which age group uses screens the most?
- What is the overall screen time distribution?

- Are there differences between males and females?
- What are the most commonly used devices?

4.2 Layer 2 – Behaviour Analysis (Why is it happening?)

Page 2 explains:

- Which devices are dominant on weekends?
- Is recreational usage higher during weekends?
- How education-to-recreation ratio changes across age bands and devices.

4.3 Layer 3 – Deep-Dive (Who are the critical cohorts?)

Page 3 highlights:

- High-risk cohorts
- Segment-based performance
- Device × Age × Location combinations
- Heatmaps identifying peak usage pockets

This layered approach allows viewers to navigate from general insights to deeper analytics.

5. Page-by-Page Dashboard Preparation (Step-by-Step)

Below is the **technical and conceptual process** of building each dashboard page in Power BI.

PAGE 1 – Exploratory Analysis Dashboard

Purpose

To give a quick understanding of how screen time varies across:

- Age bands
- Genders
- Devices

- Locations

Visuals Included

1. KPI Cards

- Average Daily Screen Time
- % of Children Exceeding Recommended Limit
- Average Educational to Recreational Ratio

Reason: These KPIs give an instant overview.

2. Histogram – Daily Screen Time Distribution

- Shows whether screen time skews towards high usage.
- Helps identify overuse patterns.

3. Bar Chart – Device Usage Count

- Displays the frequency of devices (Smartphone, Laptop, TV, Tablet).
- Useful to understand device dominance.

4. Bar Chart – Screen Time by Gender

- Compare male vs female average usage.

5. Column Chart – Screen Time by Age Band

- Highlights age groups at risk (e.g., Teens).

6. Slicer Filters (Interactive Controls)

- Gender
- Age Band
- Device Type
- Urban/Rural

Storytelling

The page tells:

“Which groups use devices the most, and how does screen-time distribution shift across demographics?”

Page 1 acts as a basic data health check and context-setting page for further analysis.

PAGE 2 – Device, Activity & Weekday/Weekend Behaviour

Purpose

To show **how usage patterns change based on activities, devices, and calendar behaviour.**

Visuals Included

1. Stacked Bar – Device Mix Across Age Bands

- Shows how device preference varies with age.

2. Clustered Bar – Device Usage (Urban vs Rural)

- Identifies technology availability patterns.

3. Line/Bar Combo – Weekday vs Weekend Screen Time

- Shows spikes on weekends.

4. Box Plot – Education/Recreation Ratio by Device Type

- Which devices encourage more educational vs recreational usage.

5. Scatter Plot – Screen Time vs Educational Ratio

- Identifies at-risk groups (high screen time + low learning ratio).

Storytelling

This page answers:

“How do habits change on weekdays vs weekends, and how does device choice influence educational engagement?”

By visualizing these relationships, we can identify when and how overuse happens.

PAGE 3 – Cohort Segmentation & Deep Insights

Purpose

To identify **high-risk cohorts** and **segments** with deep-dives.

Visuals Included

- 1. Heatmap – Age Band × Device Type (Avg Screen Time)**
 - Shows intensity zones where screen time is highest.
- 2. Heatmap – Age Band × Location (Urban/Rural)**
 - Highlights differences in screen behavior.
- 3. Top 10 Cohorts Chart**
 - Identifies specific combinations (e.g., “Urban Teens + Smartphone”).
- 4. Deviation Plot**
 - Shows how each age band deviates from mean screen time.
- 5. Quadrant Chart – Educational Line vs Screen Time**
 - Identifies:
 - Low time + High education = Healthy group
 - High time + Low education = At-risk group

Storytelling

This page answers:

“Who are the key groups driving high screen time, and how can we target interventions?”

This is the most analytical page and valuable for decision-making.

★ Final Key Insights

High-Impact Insights:

1. **Teens are the highest screen-time consumers** (especially 12–18).
2. **Smartphones dominate usage**, especially in urban settings.
3. **Weekend usage spikes by 30–50%**, proving habit-based behaviour.
4. **Educational activity decreases as screen hours increase.**
5. **Urban students show higher screen dependency** compared to rural.
6. **High-risk cohort identified: Urban Teen Smartphone Users**
7. **Age × Device × Location heatmap reveals strong intensity clusters.**
8. **Device mix changes with age** — younger kids prefer Tablets/TV, older prefer Smartphones.
9. **Screen level (Low/Med/High) aligns strongly with device type & age.**